

FumiBot: Fumigate the Future

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Abstract

In this present time, people are having trouble finding the correct cleaning materials to clean and sanitize spaces that are difficult to reach or out of sight.¹ In this period of time where outbreaks of diseases are frequent, caused by viruses and bacteria, people have become more careful of their surroundings to ensure their safety and health.² Here comes the FumiBot. FumiBot is easy to use and allows users to manually maneuver the robot from far distances using an Android phone.³ This robot can also move automatically just by pressing the object avoidance mode.³ The robot uses a sensor to check its distance from ventilation shaft walls and turns in a different direction when nearby to avoid damaging itself.³ Because of these features, users can avoid being in the same room during disinfection.³ And lastly, it has an on-off button, allowing users to activate or deactivate the robot. Upon purchase, it comes with clear instructions and labels to guide users on how to use FumiBot. This bot is essential in avoiding the spread of infectious diseases and improving wellbeing in a community since it provides a preventive strategy to ensure the safety of people.^{4,5} Its simple-to-use

design and powerful disinfecting abilities highlight its value, making it a useful tool for home as well as workplace use.^{4,6} FumiBot is a robot designed to help the community disinfect or fumigate ventilation shafts.⁷ The team's robotic solution is a step forward in keeping spaces clean—a type of innovation where ambition meets hygiene.^{6,7,8}

Keywords: *Fumigate, Ventilation Shafts, Biodegradable chemicals*

1 Introduction

Introducing FumiBot, the innovative disinfectant robot designed to ensure hygiene and safety in response to emerging diseases like COVID-19.^{9,10} With its user-friendly interface and advanced features, FumiBot offers convenient and effective disinfection for individuals and the community. FumiBot carries the idea of creating a disinfecting robot that can get rid of dirt and prevent the growth of bacteria and viruses.⁸ Individuals spend most of their time interacting with numerous people and doing different activities outside,

which may cause them to bring home unwanted germs at the end of the day.⁴ With the use of FumiBot, the bacteria and viruses causing illnesses inside workplaces, bedrooms, and other personal spaces that people frequently stay in can be minimized.^{6,7}

1.1 Background

FumiBot is a revolutionary public health technology that uses advanced robotics to fight infectious diseases like COVID-19.^{3,10} It targets surfaces that are highly touched in areas with lots of traffic and runs autonomously thanks to its advanced technology and disinfection sprays.^{3,11,3,12} By using it in places like schools, hospitals, and airports, the risk of the spread of viruses is greatly decreased.^{7,13} FumiBot makes sure that everyone is in a safer environment by doing away with the requirement for human involvement in the disinfection process.³ FumiBot is also an environmentally friendly disinfection solution that uses biodegradable chemicals.^{14,15,16} Its real-time data collecting represents a change toward intelligent, long-lasting public health solutions by improving tracking public health and operational efficiency. In addition to disinfection, FumiBot has features like temperature scanning and air quality monitoring. It can be used in any kind of circumstances, including shops and commercial buildings, thanks to its adaptability. To keep it at the forefront of public health technology, ongoing research attempts to improve its capabilities even more.

1.2 Objectives

The main objective of the team is to provide support to the community through the innovative FumiBot, an intelligent bot that has a purpose to efficiently disinfect spaces and eliminate harmful substances. Specifically,

- To fumigate ventilation shafts.
- To kill harmful microorganisms on surfaces, especially the ones people frequently touch.
- To disinfect contaminated areas that hazardous substances have been exposed to.
- To lessen the people who are getting infected by diseases.
- To prevent the growth and spread of germs, bacteria, and viruses in the homes of the consumers.
- To maintain the cleanliness of the surroundings of homes and the community.
- To ensure the safety of everyone who uses the bot.
- To be able to showcase this generation's technology capabilities and how it can become of great help towards the community.

1.3 Definitions

Below are some of the operational definition we used that ensures a shared understanding of key terms and concepts throughout the proposal:

Fumigate: One of the objectives of the team's bot is to fumigate or disinfect areas where people often go that may be contaminated by germs and bacteria. Thus, fumigate means to eliminate pests or disinfect an area by using chemicals to kill or drive away insects, rodents, or other unwanted organisms.

Ventilation shafts: As said earlier, FumiBot aims to disinfect areas where people often go; that is why the bot intends to fumigate ventilation shafts. These are vertical or inclined passages designed to facilitate the movement of air within a structure, allowing for the exchange of stale or polluted indoor air with fresh outdoor air, thereby improving air quality and regulating temperature and humidity levels.

Biodegradable chemicals: FumiBot is an environmentally friendly disinfection solution that uses biodegradable chemicals. These biodegradable substance can be chemically decomposed and broken down to simpler components by natural biological processes.

2 Project Design

2.2 Sensor Integration

The FumiBot has a sensor, which is an ultrasonic sensor.³ The ultrasonic sensor measures the distance of obstacles from the front of the robot.³ The ultrasonic sensor is the only sensor found in the FumiBot. It is attached to the front of the bot to serve as a guide to prevent it from crashing into objects.³ It measures the distance of the object from the FumiBot to the object.³ The robot would stop moving if the object is approximately 40 cm away from the front of the FumiBot.³ Object avoidance mode may be used to move automatically and follow these procedures.³

The sensor serves as a guide to prevent damage; if the object avoidance mode is active, the ultrasonic will detect it, and if it detects obstacles, it will automatically turn or steer away from them, resulting in no damage to the surroundings or the FumiBot.³

3 Methodology

3.1 Methods and approach

To ensure the success of the bot, we will be adopting and following various methods and approaches to successfully achieve the group's plan in creating the bot. Some of the important methods and approaches that we will use to guarantee the success of the bot are the following:

Gathering

The project managers, along with the project designers, will start to buy the materials needed in order to build the FumiBot using the budget of the group.^{21,22,23} Other members, like the document managers, will also help when buying and gathering the materials needed to build FumiBot.^{21,22,23} All members, mainly the Quality Assurance Team, must ensure that the materials and components needed to build FumiBot are completely bought before starting to assemble.^{21,22,23} This is to avoid situations where some of the materials needed for the proposal are missing. In addition, the project managers can think of a solution for the missing materials so that the group may proceed to the next step.

Assembling

Since the project designers are the ones who know where the right placement of the components used for the FumiBot is, we will need their help upon assembling. This is to ensure that no wire or component will get tangled or wet during the disinfection process.²² Lastly, project managers must ensure again that the manual is followed orderly and carefully so as to execute the project design properly.²⁴

Building

The FumiBot must and will be created and built perfectly to attain most of the functions of the bot. The Arduino Uno is the main element in building FumiBot since it enables the robot to function properly.²⁵ The team ensures that the procedure for the FumiBot will be correctly placed and well constructed for each component of the bot. These tackles connect the jumping wires, Arduino modules, and units in the right place for the robot to function well and properly.^{21,22,23} Additionally, for the movement, the group will be attaching four wheels to the wheelbase.²¹

For the process of making the disinfecting chemical, the group will utilize the water pump, water, container, tubes, and disinfecting chemicals.^{14,15,16}

The initial liquid will begin at the container where the water is placed.^{14,15,16} Afterwards, the water will flow in the tube, proceeding to the water pump to make the water reach the second point, where the chemicals of the disinfectant are ready to be mixed with the water.^{14,15,16} The disinfectant liquid will pass through the hose and proceed to the (copper) tube where it would spray out through the holes.^{14,15,16}

Coding

Arduino C++ is a specialized version of the C++ programming language that features a distinctive set of additional functions and libraries that expand its capabilities beyond those of standard C++.²⁶ These added functions and libraries enable developers to interact with hardware devices and sensors easily, granting them the ability to build custom electronic systems.²⁶ This language is widely supported, with many tutorials to help users learn and troubleshoot.²⁶ Lastly, the Arduino software can be used on multiple operating systems, making it easy to develop and upload code to the board.²⁶

There are multiple portions of the FumiBot that need programming.²⁶ The first portion, for the sensor, includes the ultrasonic sensor.³ The ultrasonic sensor measures the distance of obstacles from the front of the robot.³ The second portion, for the movement, includes the wheelbase.²¹ The wheelbase can be controlled using an app that can be installed on smartphones.²¹ The last portion, for spraying disinfectants, includes the water pump.^{14, 15, 16} The signal from the ultrasonic sensor will turn the water pump on or off.³

Enhancing

The project managers will double-check the bot overall to make sure that it's working well and properly. This is to ensure that there are no skipped steps in the process of building and coding.²⁴ In addition, the project managers can improve the quality of the robot with this method.

4 Budget Estimates

To Finance the purchase of FumiBot, the treasurer will be sourcing funds from members within the group. Each member is required to contribute P250, and payment is expected from all members around January 1-2. This collective effort ensures FumiBot's successful purchase for the entire group's benefit. The benefits will not only be for the group, but also for future projects and innovations. By using the funds to take FumiBot into action, the team will be able to inspire others to be able to do the same for future generations to come.

Component for the Robot and Function	Quantity	Price	Total Price
Omni Directional Mecanum Wheels Robot Car Kit Function: enables the movement of the robot Includes: 4 Hobby Gearmotor 4 Wheels 1 R3 CH340 Expanding Board 1 R3 CH340 Board 1 Bluetooth Module 1 Ultrasonic Sensor 1 Battery Case Female to Female Jumper Wires	1	P1,899.00	P1,899.00
Disinfectant Storage Function: stores the disinfectant liquid	1	N/A	N/A
Water Pump Function: controls the flow rate of the disinfectant	1	P59.00 (P58.00 Shipping Fee)	P117.00
Wireless Camera Function: allows the user to check the front view of the robot	1	P369.00 (P79.00 Shipping Fee)	P448.00

3.7V Battery Function: power the components in the robot	2	P57.00	P114.00
#22 Auto Wire (2m) Function: extend wires	1	P10.00	P10.00
TOTAL: P2,588.00			

Table 4.1 Budget Plan

5 Related Studies

For our Related studies we will show you that our Fumibot's purpose and success of being made was based on the researches of the people who research in the past that made our hypothesis that it is possible for us to make a robot that has a purpose is to provide support to the community for it to be more safe and to the people living there.

5.1 Environmental Sustainability? Goals with Examples.

Environmental preservation and protection has been a long-sought battle since the earliest of times.²⁹ As the world transitions into a new era, human activity becomes reliant on natural resources to the extent of overconsumption and degradation.²⁹ Hence, Patterson (2024) stressed the need for collective efforts toward environmental sustainability in order to preserve the intrinsic assets of Earth.²⁹ However, the massive breakthroughs of today's society made it inevitable for water, land, and air to be contaminated with different chemicals, resulting in dilemmas that traverse environmental conservation as this activity can lead to degenerations in human hygiene as these contaminated solutions may come into contact with bodies of waters that human beings utilize for everyday use, such as for drinking, and everyday chores.²⁹

Patterson's research highlights the urgent need for collective action to safeguard our environment.²⁹ However, with the rise of technology, bots can play

a crucial role in monitoring and managing natural resources more efficiently.²⁹ Fumibot can help prevent overuse and pollution, which are major threats identified by Patterson.²⁹ Still, while bots offer promising solutions, we must ensure they align with the broader goal of environmental sustainability and don't create new problems.²⁹ Thus, integrating Patterson's findings with bot technology can offer a comprehensive approach to address environmental challenges while promoting human well-being.²⁹

5.2 Pneumatic Sensor on Cleaner Robot

This study discusses an improved technological approach that is believed to enhance the cleaning power of market-driven service robots by collectively improving its coverage performance.³⁰ Although present innovation already discussed an approach for the cause, it became apparent that conventional contacts result in collisions with the presented obstacles.³⁰ Thus, this experimental investigation by Kuo et al. (2011) strategized a comparative narrative against existing to their control — iRobot Create.³⁰ iRobot Create is a digitized development that used a soft-contact-based pneumatic sensor, which was then verified as a better approach than former existing ones.³⁰

In comparison to iRobot Create, FumiBot is altered to become a revolutionary public health technology that transcends cleanliness as it is specifically made for hygiene purposes that can promote healthiness within the community.³⁰ Thus, it is presented for various locations, specialized on surfaces that are mostly touched by individuals.³⁰ Additionally, with Fumibot, collisions of all kind, including soft contacts, because it is also built with sensory technology.³⁰

5.3 The use of a UV-C disinfection robot in the routine cleaning process: a field study in an Academic hospital.

The study by Astrid et al. (2021) revealed that the perilous issues of the community with contamination make it hard for human beings alone to combat.³¹ Since there had been widespread pathogens

in society, manual cleaning and labor were deemed insufficient.³¹ Thus, the need for technological assistance was introduced in order to completely resolve the dilemma.³⁰ With the prominence of digitized and technological society, robots became the renowned development that was seen to produce outputs that are beneficial for everyday life.³¹

In light of technological advancements, bots emerge as tools in monitoring and managing natural resources more effectively. Fumibot could help mitigate the issues by its features. Though it's crucial to ensure that these technological interventions complement Astrid's recommendations for environmental sustainability. By integrating Astrid's insights with bot technology, we can develop more strategies to tackle environmental challenges and promote the well-being of both humans and ecosystems giving the same objective for Fumibot.

5.4 Commercial Cleaning: An Automated Control

In navigating through an automated cleaner, it is important that it is accessible at all times, perhaps leaning towards a remote approach of command, especially if it is designed for household purposes.³² In the study of Murdan & Ramkissoon (2020), the authors tested the plausibility of an Arduino MEGA microcontroller in connecting an android smartphone to a smart autonomous floor cleaner.³² Through Bluetooth connections, the study concluded that autonomy above this commercialized cleaning robot is feasible, while ensuring prime functions that transcend accessibility.³²

Similarly, the presented research — FumiBot — aims to adapt the same idea of remote controlling, but with the use of Arduino Uno as the main technology to use and Arduino C++ as the coding technique.³² Additionally, this innovation will have an added feature of sensor integration.³² Which can also help the user controlling the bot.³²

5.5 Decontamination of a robot used to reprocess reusable surgical instruments. Journal of Hospital Infections.

Previous research revealed the success rates of robotic machines in decontamination.³³ As per the experimentations of Heibeyn et al. (2024), eight-hour service hours for manual working methods are insufficient to cover cleaning and decontamination procedures for medical devices that are at par with cleaning standards.³³ Additionally, these procedures are too hazardous for human beings as they pose physical health risks. Therefore, a need for robot handling ensues.³³

Heibeyn and colleagues stress the urgency of collective action to protect our environment, highlighting issues like overconsumption and degradation. Fumibot offers a promising solution, providing efficient monitoring and management of natural resources. By integrating Heibeyn's insights into bot development, we can ensure alignment with environmental sustainability goals. This synergy enables comprehensive strategies to address environmental challenges effectively. Together, Heibeyn's research and bot innovations pave the way for a greener future.

5.6 Ultraviolet disinfection robots to improve hospital cleaning: Real promise or just a gimmick?. Antimicrobial Resistance & Infection Control,

Schahawi et al. (2021) also revealed that robots are exemplary tools that offer promising results in hospital disinfection and decontamination.³⁴ However, despite these robotic notations, one can see that most of these were designed to serve the field of medicine alone.³⁴ Truly, there is a wide gap in innovations that can cover the entire community, assisting each individual in their day-to-day living by providing a safer environment with preventive measures for contamination.³⁴

Schahawi et al. highlights the critical need for collaborative action to tackle environmental challenges like pollution and resource depletion. Fumibot offers a powerful solution, enabling more effective resource management and environmental mitigation efforts. By integrating Schahawi's findings into Fumibots development, we can ensure

alignment with goals of environmental sustainability.

6 Conclusion

The project proposal that has just been provided will be of great help in preparing the future for the group's latest innovation, FumiBot - a disinfectant robot.²⁰ To conclude FumiBot's proposal, the following three properties summarize the main points of the group's objectives. First things first, FumiBot is environmentally friendly, its components are unharmed and do not affect the environment at disposal.^{14,15,16,21,22,23} Second, FumiBot is also non-toxic in all aspects, it isn't harmful and dangerous to people, especially to children, pets, and people who are prone to respiratory system diseases - making it safe for everyone's use.^{14,15,16,21,22,23} Lastly, FumiBot is also user-friendly and uncomplicated to use as the package comes with clear label instructions and manuals, easy to understand for ages 13+.³⁵ These 3 main properties confirm the positive effects that FumiBot provides, which encourages others to consume the product and promote it to the world for better convenience.

In summary, FumiBot will pave the way for a healthier and much safer future for everyone. FumiBot stands out as a friendly, non-toxic, and highly efficient disinfecting robot with its unharmed parts and eco-friendly operation.^{14,15,16,21,22,23} The FumiBot maximizes disinfection procedures and covers a wide area efficiently with its ultrasonic sensor and clever navigation.³ This creative approach promises to improve the globe by creating a much safer and healthier environment while also increasing productivity in many industries.^{1,6,8,17,3}

7 Recommendation

FOR STUDENTS AND TEACHERS

FumiBot is a new technological innovation for health and safety use for everyone.^{3,13,20} It is an important assistance tool in handling school productivity, cleanliness and comfortability.^{1,6,8,17,3} As mentioned being safe and easy to use, the teachers recommended it as a guide project and a form of

modernized education for students to study and use.^{3,20} Thorough training on its safe operation and handling should be provided. Thus, the team will include clear instructions on how to use the bot correctly, as well as safety precautions to prevent accidents or exposure to harmful chemicals. Implementing safety features such as motion sensors to ensure the bot operates only when the area is clear of people can also minimize the risk of accidents. Additionally, regular maintenance and inspection of the bot's components can help identify and address any potential safety concerns before they escalate.

FOR FUTURE RESEARCHERS

FumiBot is the cue for helpful technological renovation; a blueprint for other forms of inventions and scientific projects with valuable purposes; and an involvement or helping hand in human tasks, such as cleaning and organizing, for the advantage of people in different situations.^{20,1} It is important to implement rigorous testing protocols to ensure its effectiveness and safety. This includes comprehensive testing for various surfaces, pathogens, and environmental conditions to minimize the chance of unintended consequences or ineffective disinfection. Additionally, regularly updating the bot's software and hardware based on new research and technological advancements can enhance its performance and adaptability to evolving challenges.

FumiBot is comfortable and guaranteed safe to use.^{3,14,15,16} It is best beneficial in either at home or school, or any other spaces mostly used by people.^{1,6,8,17,20,13,7} It is not intended to harm children, pets, or mostly sick people, unless misused. FumiBot not only assists household members or business crews in cleaning areas; it also monitors the safety and sanitation in areas that are more likely to be affected by dirt, germs, and bacteria, especially hospital wards and household bedrooms.^{10,6} We recommend the importance of health and safety through new technology, as we endorse FumiBot, a gateway to a cleaner and safer environment in

the future.^{3,6,8,17,20}

FumiBot ensures that the environment stays free from contamination.^{6,7,8} Our bot is a community robot that aims to lend a hand in disinfecting spaces with the help of our users.^{6,7,8,11,13,17,18} FumiBot is guaranteed to be a good alternative for lessening labor while at the same time also providing people with optimal disinfecting results.^{17,18,19,20} Our target users are those working in public places such as restaurants, hospitals, and health centers, and mainly those who prefer staying at home and are quite opposed to cleaning by themselves.^{7,6,13,1,20} In addition to that, we would also like to focus on aspiring teenagers as our target market to influence them on the impact of innovation and technology.^{3,13,20}

8 References

1. Angelina. (2022, July 18). *Why some people find it so hard to clean?* Bond Cleaning Brisbane. <https://www.bondcleaninginbrisbane.com.au/people-find-hard-clean/>.
2. Broom, D. (2022, February 15). *6 trends that define the future of health and wellness*. World Economic Forum. <https://www.weforum.org/agenda/2022/02/megatrends-future-health-wellness-covid19/>.
3. Be Smart nothing impossible. (2018, March 26). *Arduino UNO & ESP8266 and control using smart-phone [Video]*. YouTube. <https://www.youtube.com/watch?v=2cjufbgOBYo>.
4. Centers for Disease Control and Prevention. (2016, January 7). *How Infections Spread*. Centers for Disease Control and Prevention <https://www.cdc.gov/infectioncontrol/spread/index.html>.
5. Scientific Sanitation Solutions. (2023, June 10). *Disinfectants play vital role in infection control and prevention*. News-Medical.

<https://www.news-medical.net/news/20230610/Disinfectants-play-vital-role-in-infection-control-and-prevention.aspx>.

6. Centers for Disease Control and Prevention. (2023, August 29). *When and How to Clean and Disinfect Your Home*. Centers for Disease Control and Prevention.
<https://www.cdc.gov/hygiene/cleaning/cleaning-your-home.html>.

7. Cleanipedia Team. (2023, December 6). *A guide to disinfecting public Spaces: Keeping Communities safe*. Cleanipedia ZA. Cleanipedia.
<https://www.cleanipedia.com/za/floor-and-surface-cleaning/a-guide-to-disinfecting-public-spaces-keeping-communities-safe.html>.

8. National Library of Medicine. (2020, September 23). *Cleaning, disinfecting, and sanitizing*. National Library of Medicine.
<https://medlineplus.gov/cleaningdisinfectingan dsanitizing.html>.

9. Park, A. (2022, September 15). *Why Infectious Disease Outbreaks Are Becoming So Common*. TIME.
<https://time.com/6211430/infectious-disease-outbreaks-polio-covid-19/>.

10. World Health Organization: WHO. (2023, August 9). *Coronavirus disease (COVID-19)*. World Health Organization.
<https://www.who.int/newsroom/factsheets/detail/coronavirus-disease-%28covid-19%29>.

11. UF HEALTH EMERGENCY & URGENT CARE CENTER. (2022, October 11). *The Importance of Cleaning, Disinfecting and Sanitizing*. UF HEALTH.
<https://euc.ufhealthjax.org/blog/2022/october/the-importance-of-cleaning-disinfecting-and-sani/>.

12. Dhama, K., Patel, S. K., Kumar, R., Masand, R., Rana, J., Yatoo, M. I., Tiwari, R., Sharun, K., Mohapatra, R. K., Natesan, S., Dhawan, M., Ahmad, T., Emran, T. B., Malik, Y. S., & Harapan, H. (2021, May

15). *The role of disinfectants and sanitizers during COVID-19 pandemic: advantages and deleterious effects on humans and the environment*. National Library of Medicine.
<https://link.springer.com/article/10.1007/s11356-021-14429-w>.

13. ChemicalSafetyFacts. (2024, February 14). *The Crucial Role of Disinfectant Use in Public Spaces*. Chemical Safety Facts.
<https://www.chemicalsafetyfacts.org/health-and-safety/the-crucial-role-of-disinfectant-use-in-public-spaces/>.

14. Centers for Disease Control and Prevention. (2016, September 18). *Tables and Figure: Table 2, Guideline for Disinfection and Sterilization in Health-care Facilities (2008)*. Centers for Disease Control and Prevention.
<https://www.cdc.gov/infectioncontrol/guidelines/disinfection/tables/table2.html>.

15. Mississippi State University Extension. (2024, April 9). *General characteristics of Disinfectants*. Mississippi State University Extension.
<http://extension.msstate.edu/content/general-characteristics-disinfectants>

16. Mount Sinai Selikoff Centers for Occupational Health, Bellevue/NYU Occupational & Environmental Medicine Clinic. (n.d.) *Quaternary Ammonium Compounds in Cleaning Products: Health & Safety Information for Cleaners and Supervisors*. Mount Sinai Selikoff Centers for Occupational Health.
https://www.mountsinai.org/files/MSHealth/Assets/HS/Patient-Care/Service-Areas/Occupational-Medicine/QACsInfoforWorkers_18.pdf.

17. Public Health Agency of Canada. (2023, January 30). *COVID-19: Cleaning and disinfecting*. Canada.ca.
<https://www.canada.ca/en/public-health/services/diseases/2019-novel-coronavirus-infection/prevention-risks/cleaning-disinfecting.html>.

18. World Health Organization: WHO. (2020, May

- 15). *Cleaning and disinfection of environmental surfaces in the context of COVID-19*. World Health Organization.
<https://iris.who.int/bitstream/handle/10665/332096/WHO-2019-nCoV-Disinfection-2020.1-eng.pdf?sequence=1>.
19. Centers for Disease Control and Prevention. (2022, November 22). *Everyday Household Cleaning*. Centers for Disease Control and Prevention.
<https://www.cdc.gov/hygiene/cleaning/index.html>.
20. Rubin, A. (2023, October 2). *Smart Cleaning: The Role of Technology in Modern Home Cleaning*. Hellamaid.
<https://hellamaid.ca/technology/smart-cleaning-the-role-of-technology-in-modern-home-cleaning/>
21. Shopee Philippines. (n.d.). *LAFVIN Omni Directional Mecanum Wheels Robot Car Kit for Arduino UNO R3 STEM Educational DIY Coding*. Shopee.
https://shopee.ph/LAFVIN-Omni-Directional-Mecanum-Wheels-Robot-Car-Kit-for-Arduino-UNO-R3-STEM-Educational-DIY-Coding-i.365944604.23179747272?sp_atk=e5a68090-2b3b-40df-b6a8-8510f4a85f65&xptdk=e5a68090-2b3b-40df-b6a8-8510f4a85f65.
22. Shopee Philippines. (n.d.). *Mini Submersible Pump Water Fish Tanks Aquarium Fountain DC 5V USB non-USB Type*. Shopee.
https://shopee.ph/Mini-Submersible-Pump-Water-Fish-Tanks-Aquarium-Fountain-DC-5VUSB-non-USB-Type-i.18252381.621800496?sp_atk=def85281-f5fb-4bc3-84fe-a677dd886640&xptdk=def85281-f5fb-4bc3-84fe-a677dd886640.
23. Shopee Philippines. (n.d.). *A9 Camera 1080P HD Mini Portable Camera Wireless Home Surveillance WIFI IP Security Camera connect to cellphone*. Shopee.
https://shopee.ph/A9-Camera-1080P-HD-Mini-Portable-Camera-Wireless-Home-Surveillance-WIFI-IP-Security-Camera-connect-to-cellphone-i.361983754.8513139222?sp_atk=69fade7b-3a8e-4f03-a1d2-ba7a53705eb0&xptdk=69fade7b-3a8e-4f03-a1d2-ba7a53705eb0.
24. LAFVIN. (n.d.). *Mecanum Wheels Robot Car Kit*. Google Drive.
<https://drive.google.com/drive/folders/1TY2U80u2UHyEILLITG-oQ267TShfPuGw>.
25. Arduino. (2018, February 5). *What is Arduino?* Arduino.
<https://www.arduino.cc/en/Guide/Introduction>
26. Karonji, W. (2023, May 16). *What Programming Language Does Arduino Use?* MUO.
<https://www.makeuseof.com/what-programming-language-does-arduino-use/>.
27. Kuo, C.H., Chou, H.C., & Tasi, S.Y. (2011). *Pneumatic Sensor: A Complete Coverage Improvement Approach for Robotic Cleaners*. IEEE Transactions on Instrumentation and Measurement, 60(40), 1237-1256.
<https://ieeexplore.ieee.org/document/5699392>
28. Murdan, A. & Ramkissoon, P.K. (2020). *A smart autonomous floor cleaner with an Android-based controller*. International Conference on Emerging Trends in Electrical, Electronic and Communications Engineering (ELECOM).
<https://ieeexplore.ieee.org/document/9297006>
29. Patterson, N. (2020, January 16). *What is Environmental Sustainability? Goals with Examples*. Southern New Hampshire University.
<https://www.snhu.edu/about-us/newsroom/stem/what-is-environmental-sustainability#>
30. Cimolai, N. (2020). *Environmental and decontamination issues for human coronaviruses and their potential surrogates*. Journal of Medical Virology, 92(11), 2498-2510.
<https://onlinelibrary.wiley.com/doi/10.1002/jmv.26170>.
31. Kelishadi R. (2012). *Environmental pollution:*

health effects and operational implications for pollutants removal. Journal of environmental and public health, 2012, 341637.

<https://www.hindawi.com/journals/jep/2012/341637/>.

32. Astrid, F., Beata, Z., Van den Nest, M., & Elisabeth, P (2021). *The use of a UV-C disinfection robot in the routine cleaning process: a field study in an Academic hospital*. Antimicrob Resist Infect Control 10, 84.

<https://aricjournal.biomedcentral.com/articles/10.1186/s13756-021-00945-4>.

33. Matúšová, M., Bučányová, M., & Hrušková, E. (2019). *The future of industry with collaborative robots*. MATEC Web of Conferences, 299.

https://www.matec-conferences.org/articles/mateconf/abs/2019/48/mateconf_mtem2019_02008/mateconf_mtem2019_02008.html.

34. Heibeyn, J., Witte, M., & Radermacher, K. (2024). *Decontamination of a robot used to reprocess reusable surgical instruments*. Journal of Hospital Infections, 143, 1-7.

[https://www.journalofhospitalinfection.com/article/S0195-6701\(23\)00342-0/abstract](https://www.journalofhospitalinfection.com/article/S0195-6701(23)00342-0/abstract).

34. Schahawi, M.D.E., Zingg, W., Vos, M., Humphreys, A., Cerero, L.L., Fueszl, A., Zahar, R., & Presterl, E. (2021). *Ultraviolet disinfection robots to improve hospital cleaning: Real promise or just a gimmick?*. Antimicrobial Resistance & Infection Control, 10(33),

<https://aricjournal.biomedcentral.com/articles/10.1186/s13756-020-00878-4>.

35. Co, E. (2024, April 10). *ICT Bot Manual*. Google Docs.

https://docs.google.com/document/d/1ZlxJNZDcz9BX7STkQ-QQNIXJWzG4ETD2cLupkMl6keY/editfbclid=IwAR1ZUhdr9yKTYc5gBXEnKEE5hN8jRfDOY6UDAJLrEXUgsIBh_Hpv-gTBtCQ

9 Appendices

9.1 Mechanics

9.1.1 Specification. .

Material	Requirement
Battery	3.7 Volts
Disinfectant Container	525 milliliter

9.1.2 Precaution. .

1. Keep the FumiBot far away from children, animals, and plants when hazardous- chemicals are in the container.

2. Use safety gloves when handling hazardous chemicals.

3. Carefully fill the container so that it will not spill into the circuit of the FumiBot.

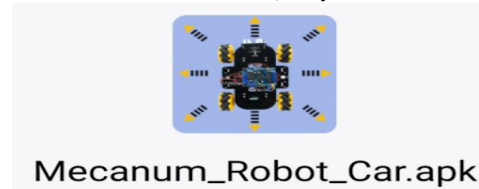
4. Regularly wash the sanitizing wipe after operations to maintain cleanliness.

5. Clean the FumiBot often, especially when hazardous chemicals were stored in the disinfectant container, to sustain the proper performance and longevity.

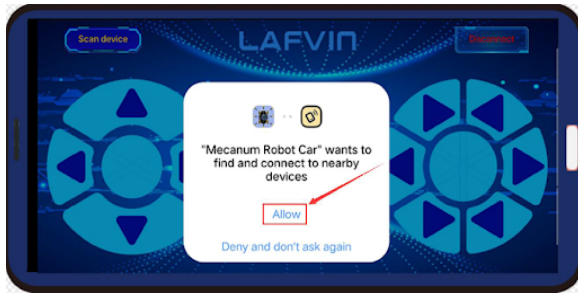
9.1.3 Installing Required Software. .

a) Bluetooth Controller App

1. Install the Mecanum_Robot_Car.apk file on the Android device (only for Android).



2. Allow the app to find and connect to nearby devices.



b) Camera App



1. Install the CY365 app to the device (only available from the Apple store for IOS and Google Play for Android).

a. Scan the APK QR code below for easy installation.

2. Allow all permissions asked by the CY365 app.



9.1.4 Operating the FumiBot. .

a) Installing the Batteries

1. Remove the cover of the battery case by sliding the cover in the direction of the arrow marked on the cover case.

2. Place the two 3.7V batteries in the battery case following the marked orientation inside.

3. Cover the battery case by sliding the cover in the opposite direction of the arrow marked on the cover case.

b) Starting the FumiBot

1. Slide the switch to ON placed on the side of the battery case.

2. Click the ON/OFF button of the camera.

3. Wait until the blue LED light next to the button blinks continuously (waiting state).

c) Connecting to the HC-06 Bluetooth Module

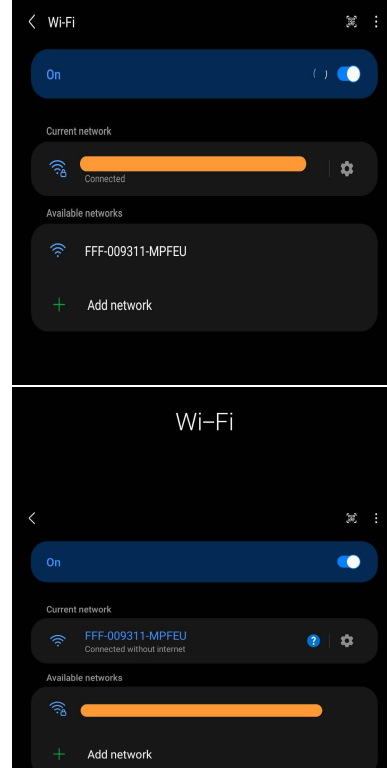
1. Turn on Bluetooth on the Android device.

2. Pair the Android device with the HC-06 Bluetooth module.

3. Enter passcode 1234.

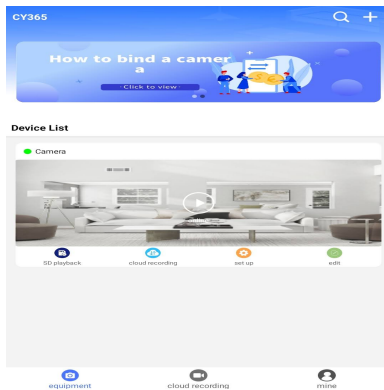
d) Connecting to the Camera

1. Turn on Bluetooth and connect the Wifi of the device to FFF-009311-MPFEU.

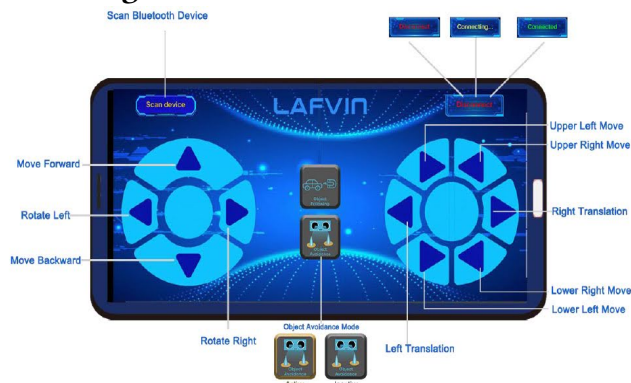


2. Open the CY365 app.

3. Search for the camera from the device list of the CY365 app, then click the play button.



e) Controlling the Bot



(the image above shows the functions of the controller)

1. Open the Bluetooth Controller App.
2. Press the “Scan Device” button.
3. Select the HC-06 device.
4. Wait for the box on the upper right side of the screen to display “Connected” while “Connecting...” is displayed.
5. Press any arrow buttons to control the FumiBot.
6. The Ultrasonic Avoidance Mode could be turned ON or OFF by pressing its button.

f) Shutting Down the FumiBot

1. Slide the switch to OFF placed on the side of the battery case.
2. Press the ON/OFF button of the camera until the blue LED light next to the button turns off.

9.1.5 Troubleshooting. .

a) Bluetooth Controller App

1. The app reports: Error 507: Unable to connect. Is the device turned on?



- The Fumibot might be off. Turn on the power switch.
- Check if the HC-06 Bluetooth module’s red LED light is flashing (available) and selected.
- If the HC-06 Bluetooth module’s LED light flashes rapidly (available status), try switching OFF and then ON the power supply.

2. If you click to search for devices, the device list does not have any selectable devices.

The app needs permission to access nearby devices.

If the HC-06 Bluetooth device cannot be found in the “Scan Device” button, Unpair and then pair the HC-06 Bluetooth module to the device.

b) Camera App

1. If the CY365 app search cannot detect the camera

> Confirm that the device has connected to the camera.

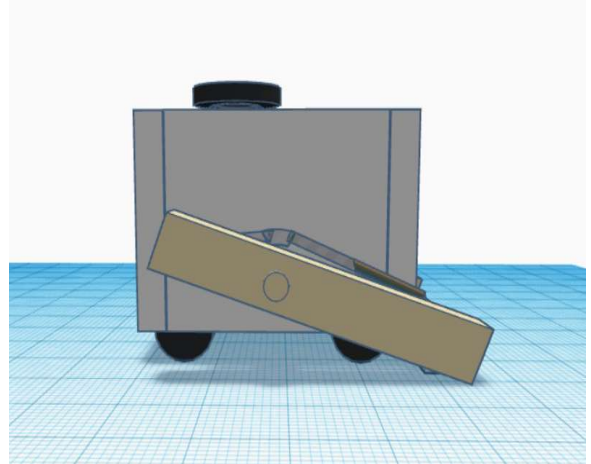
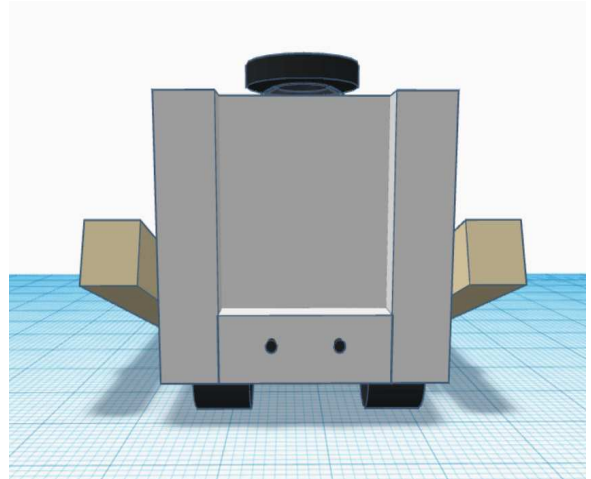
> Check whether all app permissions have been allowed.

2. If the camera cannot connect to the device

> The camera might have a low battery, so recharge the camera.

9.2 Technical Specifications

Capacity of Disinfectant Container	About 500 mL
Speed of the FumiBot	May Vary
Dimensions of FumiBot	Length (short side): around 6.5 inches Width (long side): around 8 inches Height: around
Dimensions of Wheelbase	Length (short side): around 6.5 inches Width (long side): around 8 inches Height (from wheels): around 4.5 inches
Capacity of Disinfectant Container	About 500 mL



9.3 Project Design

(FumiBot 3D Model)

